

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.1 REAL NUMBERS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished the unit on real numbers. Now write your own Final Explanation to answer the Driving Question: Why can some numbers be written as exact fractions and others cannot, and how does knowing a number's type help us solve everyday problems in Kenya? Write your answer in the blank space under each prompt. Use evidence from your lessons: the Nairobi pizza diagrams, your fraction-circle results, your Number World Map, your reciprocal comparison table and your Driving Question Board. Include worked numbers where the prompt asks for them. Use full sentences and show your reasoning.

Part 1: The Pizza Puzzle and Whole-Number Types

Think back to the friends in Nairobi sharing a pizza. (a) Using your fraction-circle results, describe what happened to the decimal for one slice when the pizza was cut into 4 slices and when it was cut into 3 slices. (b) Whole numbers can be sorted as odd, even, prime or composite. Choose any three numbers from the set 2, 3, 4, 9, 15, 17, 36, 41, 49, 64 and say how you classified each and why. (c) Explain why a pizza cut into a prime number of slices, such as 3 or 7, is harder to share equally between 2 people than one cut into 4 or 8 slices. Mention what is special about the number 1 and the number 2.

(a) When the pizza was cut into 4 equal slices, each slice was $\frac{1}{4} = 0.25$. This decimal ends, so it is a terminating decimal. When it was cut into 3 equal slices, each slice was $\frac{1}{3} = 0.3333\dots$, a decimal that never ends but repeats the digit 3. Our fraction-circle table showed that cuts into 2, 4, 5, 8 and 10 gave decimals that end, while cuts into 3, 6 and 7 gave decimals that repeat or go on and on. (b) Example classifications: 17 is odd and prime because its only factors are 1 and 17. 36 is even and composite because it has many factors (1, 2, 3, 4, 6, 9, 12, 18, 36). 49 is odd and composite because it has three factors: 1, 7 and 49. (c) A prime number of slices has only two factors, 1 and itself, so the slices cannot be regrouped evenly into smaller whole-number groups. With 3 or 7 slices, sharing between 2 people means cutting at least one slice again. A composite number such as 4 or 8 can be split into equal groups, so sharing is clean. The number 1 is neither prime nor composite. The number 2 is the only number that is both even and prime.

Part 2: Rational and Irrational Numbers

(a) Explain what makes a

(a) A rational number can be written as a fraction $\frac{p}{q}$ where p and q are integers and q is not 0. Its decimal either terminates or

number rational and what makes a number irrational. Talk about p/q and about what the decimal does. (b) Classify each of these as rational or irrational and give your reason: 0.75, $1/3$, $\sqrt{2}$, π. (c) Some people use $22/7$ for π. Is $22/7$ equal to π? Explain. (d) Use your number-line strip: between which whole numbers did $\sqrt{2}$ and π sit, and what does this tell you about a number that has no exact fraction?	repeats. An irrational number cannot be written as p/q. Its decimal never terminates and never repeats. (b) 0.75 is rational because it terminates and equals $3/4$. $1/3$ is rational because it is a fraction and its decimal 0.333... repeats. $\sqrt{2} \approx 1.41421356...$ is irrational because the decimal never ends and shows no repeating block. $\pi \approx 3.14159265...$ is irrational for the same reason. (c) No. $22/7$ is a fraction of two integers, so it is a rational number, and it is only an approximation of π. π is irrational and cannot be written as an exact fraction. (d) $\sqrt{2}$ sits between 1 and 2, closer to 1.4. π sits between 3 and 4, closer to 3.14. This shows that an irrational number is a real number with a definite place on the number line, even though we can never write its exact value as a fraction or an ending decimal.
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Part 3: Reciprocals

(a) In your own words, what does the reciprocal of a number tell us in the pizza situation? (b) Find the reciprocal of $1/4$, of $2/3$ and of $\sqrt{2}$. Say whether each reciprocal is rational or irrational. (c) Does every real number have a reciprocal? Explain. (d) Amina in a mandazi problem has 1 kg of flour and uses $2/3$ kg per batch. Show how reciprocals help you find the number of batches.	(a) The reciprocal of a slice size n is $1/n$. It tells us how many slices of that size fit into the whole pizza. (b) The reciprocal of $1/4$ is 4, which is rational. The reciprocal of $2/3$ is $3/2$, which is rational. The reciprocal of $\sqrt{2}$ is $1/\sqrt{2} = \sqrt{2}/2 \approx 0.7071...$, which is irrational. (c) Every non-zero real number has a reciprocal, whether it is rational or irrational. Zero has no reciprocal because division by zero is undefined. (d) Number of batches = $1 \div 2/3 = 1 \times 3/2 = 3/2 = 1.5$ batches. Dividing by a fraction is the same as multiplying by its reciprocal, so reciprocals turn a division into a multiplication.
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Part 4: Choosing a Method and Predicting by Type

(a) Name the three methods you used to find reciprocals. (b) You found the reciprocal of $1 \div 7$ and of $1 \div \sqrt{2}$ using all three methods. What did you notice about the results for each, and why? (c) $\sqrt{4}$ is written with a radical sign. Is it rational or irrational, and what does this mean for its reciprocal? (d) Explain how knowing a	(a) The three methods are long division by hand, mathematical tables and a calculator. (b) For the rational number 7, all three methods gave consistent results for its reciprocal, $1/7$. For the irrational $\sqrt{2}$, the methods gave slightly different approximations, because the decimal never ends so every method must round. Calculators are fast and precise but still round. Division is exact for rational numbers. Tables are reliable and portable. (c) $\sqrt{4} = 2$, so it is rational. Its reciprocal $1/2$ is exact. A radical sign alone does not make a number irrational. (d) If the number is rational, its reciprocal will be rational and can be written as a neat fraction. If the number is irrational, its reciprocal will be irrational, so we can only approximate it as a decimal. Knowing the type tells us before we calculate whether an exact answer exists.
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number's type lets you predict something about its reciprocal before you calculate.	
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Part 5: Final Explanation and Kenyan Transfer	
Answer the Driving Question in a connected explanation of 6 to 8 sentences: Why can some numbers be written as exact fractions and others cannot, and how does knowing a number's type help us solve everyday problems in Kenya? Use the Nairobi pizza, at least one Kenyan context from your lessons (for example mandazi in Kisumu, fertiliser bags in Eldoret, tiles in Mombasa, sharing land in the Rift Valley or githeri portions) and at least two key terms from this unit.	Some numbers can be written as exact fractions because they come from dividing a whole into a finite number of equal parts, so they have the form p/q and their decimals either end or repeat. These are rational numbers, like the $1/4$ and $1/3$ slices of the Nairobi pizza. Other numbers, such as $\sqrt{2}$ and π, are irrational: their decimals never end and never repeat, so no fraction can capture them exactly. A slice whose size involves an arc length with $\sqrt{2}$ or π really exists on the pizza, but we can only approximate its size. The reciprocal $1/n$ tells us how many such slices fit into the whole, and it works for every non-zero real number. The reciprocal of a rational number is rational and the reciprocal of an irrational number is irrational. This helps in Kenya: when Amina in Kisumu divides 1 kg of flour by $2/3$ kg per batch, the exact answer is 1.5 batches because both numbers are rational. When dividing githeri portions or land in the Rift Valley, a rational measure gives an exact fair share, but an irrational measure such as a tile diagonal in Mombasa can only be approximated, so we choose division, tables or a calculator according to the precision we need.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification of numbers (whole, rational, irrational)	Correctly classifies all numbers, gives accurate definitions with p/q and decimal behaviour, and corrects the $22/7$ and $\sqrt{4}$ ideas.	Classifies most numbers correctly with mostly accurate definitions; may miss one subtle case such as $22/7$ or $\sqrt{4}$.	Confuses types or gives incomplete definitions with few reasons.
Reciprocals and calculation	Finds all reciprocals correctly, explains what a reciprocal means, notes zero has none, and solves the mandazi problem with clear steps.	Finds most reciprocals correctly and explains the meaning, with small gaps in steps or in the zero case.	Makes errors in reciprocals or cannot explain what they mean.
Use of evidence from the unit	Uses specific evidence such as fraction-circle decimals, number-line positions and method comparisons to support claims.	Uses some evidence from the unit but it is general in places.	Little or no reference to evidence from lessons.
Connecting number type to prediction and the phenomenon	Clearly explains how type predicts whether a reciprocal or slice is exact or approximate and links it to the pizza and	Links type to exact or approximate answers, with a partial connection to the pizza or a Kenyan context.	Does not connect number type to prediction or to the phenomenon.

	Kenyan contexts.		
Mathematical communication	Writes clear, well-organised sentences using correct terms such as rational, irrational, terminating, repeating and reciprocal.	Writing is mostly clear with generally correct terms.	Writing is unclear or uses terms incorrectly.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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